Indian Institute of Science Education and Research
Bhopal

Deep Learning (DSE316/616)

Assignment–2

Deadline: 20-03-2026, 11:59 PM | Max Marks: 10 + 10 (Bonus)

Please follow the instructions carefully

1. Any form of plagiarism or copying (including similar submissions) will be treated as academic dishonesty and may result in being barred from the course.
2. All models must be implemented in **Python**. You may use standard deep learning libraries such as PyTorch or TensorFlow.
3. Clearly comment your code to explain the architecture, training process, and design choices.
4. You are free to use AI toolboxes with proper references of the toolbox, library, online sources, and knowledge about what exactly the code is doing. If you can explain your code, we are happy with your AI codes.
5. Submit:
   - Properly formatted and commented source code.
   - A detailed report (PDF).
   - A brief (1-2 slides) presentation summarizing your analysis.

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1. **Train each deep network including VGG, ResNet, ConvNeXT, and ViT using CIFAR-10, F-MNIST, and ImageNet-100 datasets.**

   In this assignment, if no validation set is provided, use 20% of the training data points as the validation set. Keep them separate in the beginning itself so that they are the same across experiments.

   - Corruption Library: `https://github.com/bethgelab/imagecorruptions`
   - Corrupted Datasets: `https://github.com/hendrycks/robustness`
   - Datasets: (i) `https://github.com/fendy07/FMNIST-DeepLearning`, `https://www.cs.toronto.edu/~kriz/cifar.html`, `https://huggingface.co/datasets/clane9/imagenet-100`, `https://www.kaggle.com/datasets/ambityga/imagenet100`
   - Train the same MLP architecture using the following form of validation set:
     - When the validation set is clean. (2 Marks)
     - When the validation set (not training set) is corrupted. For corruption, feel free to use any library such as the one given above. (4 Marks)

  - Optimize the amount of corruption needed to corrupt the validation set. In other words, should we perturb the validation set using all corruption (say 15 available in the above library). If you want you can fix the corruption severity to 2. (4 Marks)
  - [**Optional**] Instead of corrupting the validation, what if we perturb the intermediate feature map. You can experiment with what forms of layer play what role in robustness (first few layers, middle layers, last few layers or combination of them). (10 Marks)

- [**Must be done, No marks without this in any of the above components**] (i) Report the classification accuracy on the clean and corrupted test set of each dataset and see how the accuracy and robustness (accuracy on corrupted images) of the models are changing with different training and validation distributions (clean vs. corrupted). (ii) Visualize the change in the decision boundary. The probable ways can be looking at the feature maps against each setting, t-SNE plots showing the separation learned between classes, etc. This must be there with each training setting mentioned above.